Σ_{E} or, more generally, a distribution. For a canonical system there is no longer a constraint of the form $H = \mathrm{E}$ and thus we must allow the phase space to include essentially all points for which the position coordinates lie in the defined configurational domain (a 3-torus, if periodic boundary conditions are used). In general the momenta are unbounded. It thus becomes necessary to restrict the set of functions on which the functional $\langle \cdot, \rho \rangle$ acts. In the case of periodic boundary conditions we will assume that, for any observable of interest,

$$|\phi(\boldsymbol{q},\boldsymbol{p})| = \mathcal{O}(\|\boldsymbol{p}\|^{\alpha}), \qquad \boldsymbol{p} \to \infty,$$

for some positive constant α.

If the position domain $\mathcal{Q}$ is unbounded (for example choosing $\mathcal{Q} = \mathbb{R}^{\mathrm{N_c}}$) then some other restriction is needed on the potential energy function U (and on the observables) in order that averages are well defined. Typical assumptions for potentials on the unbounded domain are that (i) U is bounded below, and (ii) U grows sufficiently rapidly as $\boldsymbol{q} \to \infty$. This ensures that

$$\int_{\mathcal{Q}} \exp(-\beta U(\boldsymbol{q}))\mathrm{d}\omega_q < \infty.$$

Under assumptions on the potential energy as discussed above, and with ϕ continuous and allowed to grow at most polynomially fast at infinity, we may proceed to define averages with respect to the canonical density ρ_β, namely

$$\mathrm{Av}_\beta(\phi) = Z_{\mathrm{can}}^{-1} \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega. \tag{6.4}$$

6.1.5 Calculation of Temperature

Consider the one degree-of-freedom model $H(q,p) = \frac{p^2}{2m} + U(q)$ on the unbounded domain, with U bounded below and growing sufficiently rapidly as $|q| \to \infty$. The associated canonical density is

$$\rho_\beta = Z_{\mathrm{can}}^{-1} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right),$$

where

$$Z_{\mathrm{can}}^{-1} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \exp\left(-\beta\left[\frac{p^2}{2m} + U(q)\right]\right)\mathrm{d}p\mathrm{d}q.$$

We will denote the average with respect to the canonical probability density by Av_β. Using (6.4), we can compute the average of the function p^2 as

$$\begin{aligned}\mathrm{Av}_\beta(p^2) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p^2 \rho_\beta \mathrm{d}q\mathrm{d}p \\ &= \frac{\int_{-\infty}^{\infty} p^2 \exp(-\beta m^{-1}p^2/2)\mathrm{d}p}{\int_{-\infty}^{\infty} \exp(-\beta m^{-1}p^2/2)\mathrm{d}p},\end{aligned}$$

where we have used the fact that the exponential of H separates into a product of p and q dependent factors. We then integrate by parts to obtain

$$\mathrm{Av}_\beta(p^2) = m\beta^{-1},$$

which means that

$$\mathrm{T} = \mathrm{Av}_\beta\left(\frac{p^2}{m\mathrm{k_B}}\right).$$

Thus the temperature can be related to the canonical average of the kinetic energy.

If we consider a $\mathrm{N_d} = \mathrm{N_c}$ degree of freedom system with kinetic energy $K = \sum_{i=1}^{\mathrm{N_c}} p_i^2/(2m_i)$ and potential energy $U(q_1, q_2, \ldots, q_{\mathrm{N_c}})$, then we may decompose the unnormalized canonical distribution as

$$\begin{aligned}\exp\left(-\beta[K(\boldsymbol{p}) + U(\boldsymbol{q})]\right) &= \exp\left(-\beta K(\boldsymbol{p})\right)\exp\left(-\beta U(\boldsymbol{q})\right) \\ &= \exp\left(-\beta U(\boldsymbol{q})\right)\prod_{i=1}^{\mathrm{N_c}}\exp\left(-\beta\frac{p_i^2}{2m_i}\right).\end{aligned}$$

Then we again find, by the integration argument given above, that

$$\mathrm{Av}_\beta(m_i^{-1}\mathrm{k_B}^{-1}p_i^2) = \mathrm{T}.$$

Thus the average kinetic energy of any individual particle is directly related to the temperature, and thus also

$$\mathrm{Av}_\beta(2K) = \mathrm{N_c}\mathrm{k_B}\mathrm{T}.$$

Very often, this is taken to be the definition of temperature in practical settings. However, since the temperature relation is derived by integration with respect to